# multiSpeciesRegionFoam

Physics Reference Guide

Species transport through solid membranes and porous materials
in the OpenFOAM-13 `foamMultiRun` framework

**Scope.** This document describes every physical model implemented in the `multiSpeciesRegionFoam` library: governing equations, material constitutive laws, interface and surface boundary conditions, performance metrics, and numerical discretisation. Each section indicates which tutorial cases (case311–case322) demonstrate the corresponding physics.

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## Introduction and Scope

`multiSpeciesRegionFoam` is an OpenFOAM-13 add-on that extends the native `foamMultiRun` [7] multi-region framework with coupled species transport. Each solid region runs the `speciesSolid` solver module, which adds a species conservation equation alongside the inherited thermal energy equation. Inter-region coupling (heat and species) uses OpenFOAM's mapped-patch infrastructure, and all new boundary conditions are registered in the standard run-time selection tables [1].

**Physical scope.** The library targets *diffusion-dominated* transport of a dilute dissolved species (hydrogen isotopes H/D/T, vapour, dissolved salt) through one or more solid layers. Convection inside the solid and binary diffusion effects are neglected; the driving force is the concentration gradient in each layer. Typical applications are:

- Hydrogen permeation through hydrogen carrier pipes and storage tanks
- Tritium permeation through first-wall and breeding blanket materials in fusion machines, and through structural materials in molten salt reactors (H, D, T transport; Sieverts-law solubility).
- Dense-film membrane separation: reverse osmosis, membrane distillation, pervaporation (Henry-law solubility, Arrhenius temperature dependence).
- Permeation barriers (tungsten carbide, alumina coatings) [4].
- Hydrogen permeation in heat exchangers with molten-salt coolants.

## Governing Equations

### Species conservation

In each solid region the *mobile* species concentration $C\,[\mathrm{mol\ m^{-3}}]$ satisfies:

**Species equation — every solid region**

$$\frac{\partial C}{\partial t} = \nabla \cdot (D(T)\,\nabla C) - \frac{\partial C_t}{\partial t} + S_{\mathrm{vol}} \tag{1}$$

| Symbol | Meaning | Units |
|---|---|---|
| $C$ | Mobile species concentration | $\mathrm{mol\ m^{-3}}$ |
| $D(T)$ | Temperature-dependent diffusivity | $\mathrm{m^2\ s^{-1}}$ |
| $C_t$ | Trapped concentration (§3.3) | $\mathrm{mol\ m^{-3}}$ |
| $S_{\mathrm{vol}}$ | Uniform volumetric source | $\mathrm{mol\ m^{-3}\ s^{-1}}$ |

When trapping is disabled (`trappingModel noTrapping`), $\partial C_t/\partial t = 0$ and the equation reduces to pure diffusion. The source term $S_{\mathrm{vol}}$ is optional and uniform within each region.

**Demonstrated by:** all twelve tutorial cases. Cases `case311`–`case312` use the thermal field $T$ as a species proxy to verify the pure-diffusion solver before species coupling is introduced.

### Energy conservation

The thermal energy equation is inherited from OpenFOAM-13's `solid` module:

**Energy equation (Fourier conduction)**

$$\frac{\partial(\rho\, e)}{\partial t} + \nabla\cdot\mathbf{q} = S_e, \qquad \mathbf{q} = -\kappa(T)\,\nabla T \tag{2}$$

| Symbol | Meaning | Units |
|---|---|---|
| $\rho$ | Density | kg m$^{-3}$ |
| $e$ | Specific internal energy | J kg$^{-1}$ |
| $\mathbf{q}$ | Conductive heat flux | W m$^{-2}$ |
| $\kappa(T)$ | Thermal conductivity | W m$^{-1}$ K$^{-1}$ |
| $S_e$ | Volumetric heat source | W m$^{-3}$ |

The species and energy equations are solved *sequentially* within each PIMPLE outer corrector. After the energy equation updates $T$, the diffusivity $D(T)$ (and all Arrhenius properties) are refreshed before assembling the species equation.

**Full energy–species coupling:** `case316` (membrane distillation with Arrhenius $D(T)$ driven by a temperature gradient), `case320` (three-region shell-and-tube heat exchanger; conjugate heat transfer through FLiBe–SS316–Hitec).
**Isothermal (energy solved but $T$ fixed):** `case311`–`case315`, `case317`–`case319`, `case321`–`case322`.

### Species and energy coupling

In multi-region cases, the PIMPLE loop exchanges temperature across coupled patches via `coupledTemperature` boundary conditions (standard OpenFOAM), and exchanges species concentration via the custom `sievertsCoupledMixed` or `surfaceRecombination` BCs described in §4. Both exchanges happen every PIMPLE outer corrector, so the fields remain thermodynamically consistent throughout the time step.

## Material Models

### Arrhenius temperature dependence

Every material property that depends on temperature follows the Arrhenius form. This applies uniformly to $D$, the solubility $K_s$, and the surface-kinetics coefficients $K_d$, $K_r$:

**Arrhenius property**

$$X(T) = X_0 \exp\left(-\frac{E_a}{RT}\right) \tag{3}$$

| Symbol | Meaning | Notes |
|---|---|---|
| $X_0$ | Pre-exponential factor | Carries the SI units of $X$ |
| $E_a$ | Activation energy | J mol$^{-1}$ |
| $R$ | Gas constant | 8.314 J mol$^{-1}$ K$^{-1}$ |
| $T$ | Local temperature | K |

Setting $E_a = 0$ gives a temperature-independent constant, used in all isothermal tutorial cases.

**Dictionary syntax** (applies to all Arrhenius properties):

```
D { X0  [0 2 -1 0 0 0 0]  2.9e-7;   Ea  13800; }
```

**Isothermal ($E_a = 0$):** `case311`–`case315`, `case317`–`case319`, `case321`–`case322`.
**Arrhenius $D(T)$ active:** `case316` (D(T) for vapour diffusion), `case319` (real $D(T)$ for WC and SS316), `case320` (real $D(T)$ for SS316 wall at $T$ gradient from FLiBe/Hitec).

### Solubility laws

*Sieverts' law (metals and interstitial solids)*

Sieverts' law describes the equilibrium solubility of diatomic gases ($H_2$, $D_2$, $T_2$) that dissociate upon dissolution. The dissolved concentration is proportional to the *square root* of the gas partial pressure:

**Sieverts' law**

$$C \;=\; K_s(T)\,\sqrt{p} \tag{4}$$

$K_s(T)\,[\,\mathrm{mol\ m^{-3}\ Pa^{-1/2}}]$ is the Sieverts solubility constant, following the Arrhenius form (3). Applicable materials include iron, stainless steel (SS316), tungsten, vanadium alloys, and palladium [6].

Used in: `case312`–`case314`, `case318`–`case320`, `case321` (metal layer), `case322`.

*Henry's law (polymers, liquids, porous media)*

For species that dissolve without dissociation (vapour in a polymer, dissolved gas in a liquid), the equilibrium concentration is proportional to pressure:

**Henry's law**

$$C \;=\; K_H(T)\,p \tag{5}$$

$K_H(T)\,[\,\mathrm{mol\ m^{-3}\ Pa^{-1}}]$ is the Henry constant. Applicable to dense polymer films, membrane distillation membranes, and organic solvents [5].

Used in: `case315`–`case316` (membrane distillation), `case317` (reverse osmosis, $K_H = 0.5$ at the feed/membrane interface), `case321` (polymer layer).

### McNabb–Foster trapping kinetics

In metals, hydrogen isotopes can occupy lattice defects (vacancies, dislocations, grain boundaries). The McNabb–Foster model [2] treats a single population of trap sites with density $n_t = f_t N\,[\,\mathrm{mol\ m^{-3}}]$:

**McNabb–Foster ODE**

$$\frac{\partial C_t}{\partial t} \;=\; \frac{\alpha_t}{N}\,C\,(n_t - C_t) \;-\; \alpha_d(T)\,C_t \tag{6}$$

| Symbol | Meaning | Units |
|---|---|---|
| $N$ | Molar trap-site density | mol m$^{-3}$ |
| $f_t$ | Trap fraction ($0 < f_t \leq 1$) | — |
| $n_t = f_t N$ | Available trap-site density | mol m$^{-3}$ |
| $\alpha_t$ | Trapping rate coefficient | s$^{-1}$ |
| $\alpha_d(T) = \alpha_{d0} \exp(-\varepsilon/k_B T)$ | Detrapping rate | s$^{-1}$ |
| $\varepsilon$ | Trap binding energy | eV |
| $k_B$ | Boltzmann constant | J K$^{-1}$ |

The trapping term $\partial C_t/\partial t$ is subtracted from the mobile species equation (1) as a sink.

*Trapping regimes*

Two limiting regimes exist depending on the ratio of trapping to diffusion time scales:

- **Weak trapping** ($\alpha_t/D \ll 1$): traps fill slowly; the mobile front propagates at nearly the bare diffusion speed; breakthrough time $\approx L^2/D$.
- **Strong trapping** ($\alpha_t/D \gg 1$): most atoms are trapped; effective diffusivity $D_{\text{eff}} = D/(1 + N_t\,\alpha_t)$ is greatly reduced; breakthrough is delayed by orders of magnitude.

*Numerical discretisation of the trapping ODE*

The ODE (6) is stiff and must be discretised carefully to remain bounded ($0 \leq C_t \leq n_t$). A per-cell semi-implicit Euler step is used:

**Semi-implicit Euler for trapping ODE**

$$C_t^{n+1} = \frac{C_t^n + \Delta t\,\dfrac{\alpha_t}{N}\,C\,n_t}{1 + \Delta t\left(\dfrac{\alpha_t}{N}\,C + \alpha_d\right)} \tag{7}$$

This scheme is unconditionally stable and guarantees $0 \leq C_t^{n+1} \leq n_t$ for any time step.

**Demonstrated by:** `case313`. Two sub-cases span the weak and strong trapping regimes. The analytical breakthrough time is used to validate both limits.

## Interface and Surface Boundary Conditions

### Overview

At every coupled patch, two physical conditions must hold simultaneously:

1. **Flux continuity:** $D_A\,(\partial C_A/\partial n)_{\text{face}} = D_B\,(\partial C_B/\partial n)_{\text{face}}$
2. **Thermodynamic equilibrium:** determined by the solubility law on each side (§3.2).

These are implemented as a *mixed* (Robin) boundary condition in OpenFOAM, derived from the conjugate heat-transfer analogy.

### Sieverts–Sieverts interface (linear partition)

When both sides follow Sieverts' law, the equilibrium condition at the interface face is:

**Sieverts–Sieverts partition (`partition linear`)**

$$\frac{C_A}{K_{s,A}} = \frac{C_B}{K_{s,B}} \qquad \Longleftrightarrow \qquad C_A = \frac{K_{s,A}}{K_{s,B}} C_B \tag{8}$$

The concentration ratio $K_{s,A}/K_{s,B}$ can be greater or less than unity, producing a concentration jump at the interface even in steady state. When $K_{s,A} = K_{s,B}$, the concentration is continuous.

**Mixed-BC coefficients** (self = region $A$, neighbour = region $B$):

$$\begin{aligned} \text{refValue} &= \frac{K_{s,A}}{K_{s,B}} C_{B,\text{cell}} \\ w &= \frac{D_B\,\delta_B^{-1}}{D_B\,\delta_B^{-1} + D_A\,\delta_A^{-1}(K_{s,A}/K_{s,B})} \\ C_{A,\text{face}} &= w \cdot \text{refValue} + (1-w)\,C_{A,\text{cell}} \end{aligned} \tag{9}$$

where $\delta^{-1}$ is the distance from the cell centre to the interface face.

**Demonstrated by:** `case314` (two-layer composite membrane, $K_s$ ratio = 4; interface concentration jump of factor 4 at steady state), `case318` (code-to-code benchmark, equal $K_s$, continuous profile), `case319` (WC/SS316 barrier, large $K_s$ ratio responsible for PRF), `case320` (SS316 wall; also `case321` reference sub-case).

### Henry–Sieverts interface (nonlinear partition)

When a Henry-law material (polymer, liquid) interfaces with a Sieverts-law material (metal), the same partial pressure $p_{\text{int}}$ must hold on both sides:

$$p_{\text{int}} = \frac{C_{\text{Henry}}}{K_H} = \left(\frac{C_{\text{Sieverts}}}{K_s}\right)^2$$

This gives two equivalent partition conditions depending on which side is considered "self":

**Henry–Sieverts partition**

$$\text{Sieverts side (`partition sqrt`):} \quad C_{\text{Sieverts}} = K_s\sqrt{\frac{C_{\text{Henry}}}{K_H}} \tag{10}$$

$$\text{Henry side (`partition quadratic`):} \quad C_{\text{Henry}} = K_H\left(\frac{C_{\text{Sieverts}}}{K_s}\right)^2 \tag{11}$$

Both sides must be set consistently in the `0/` boundary field files; they encode the same physics from opposite perspectives.

**Linearised mixed-BC weight** for the nonlinear cases. The weight (9) is derived by linearising the partition function at the current Picard iterate. For the `sqrt` mode (self = Sieverts, neighbour = Henry):

$$\left.\frac{\mathrm{d}C_s}{\mathrm{d}C_n}\right|_{\text{iterate}} = \frac{C_{s,\text{face}}}{2\,C_{n,\text{cell}}} \qquad \Rightarrow \qquad \text{selfKD} = D_A\,\delta_A^{-1}\,\frac{C_{A,\text{face}}}{2\,C_{B,\text{cell}}} \tag{12}$$

Using the linearised derivative (rather than the constant $K_s/K_H$) in the weight formula ensures that flux continuity is enforced exactly at every Picard step, even in steady state on a coarse mesh.

*Steady-state interface concentrations*

For the case of equal layer thicknesses $L$, equal diffusivities $D$, inlet concentration $C_0 = 1$ mol m$^{-3}$, and $K_s = K_H = 1$:

**Analytical interface concentrations (equal $D$, $L$, $K$)**

$$u^2 + u = 1, \quad u = C_{\text{Sieverts,int}}, \qquad \Rightarrow \qquad u = \tfrac{\sqrt{5}-1}{2} \approx 0.618 \text{ mol m}^{-3} \tag{13}$$

The *golden-ratio conjugate* $(\sqrt{5}-1)/2$ appears naturally because the steady-state flux equality combined with the $\sqrt{\cdot}$ partition gives exactly the quadratic $u^2 + u - 1 = 0$. The Henry (polymer) side holds $C_{\text{Henry,int}} = u^2 \approx 0.382$ mol m$^{-3}$, demonstrating that the *concentration is discontinuous and higher on the Sieverts (metal) side* — the defining signature of this partition law.

The steady-state flux in the Henry–Sieverts case is 23.6 % *higher* than the equivalent Sieverts–Sieverts case with the same $D$, $L$, and $K$:

$$J_{\text{HS}} = D\,\frac{C_{\text{Sieverts,int}}}{L} = \frac{(\sqrt{5}-1)/2}{L/D}\,C_0 \;\approx\; 1.236\,J_{\text{SS}} \tag{14}$$

**Demonstrated by:** `case321`. Two sub-cases (`sieverts_sieverts` reference and `henry_sieverts`) run with identical $D$, $L$, boundary values; the 23.6 % flux difference is confirmed analytically and numerically.

## Surface recombination and dissociation

At a free metal surface exposed to a molecular gas, two kinetic processes compete:

**Dissociative absorption:** $H_2 \to 2H$, flux $= K_d(T)\,p_{\text{gas}}$
**Recombinative desorption:** $2H \to H_2$, flux $= K_r(T)\,C^2$

The net flux into the solid is the *surface recombination Robin BC*:

**Surface recombination BC (`surfaceRecombination`)**

$$D\,\frac{\partial C}{\partial n} \;=\; K_d(T)\,p_{\text{gas}} \;-\; K_r(T)\,C^2 \tag{15}$$

| Symbol | Meaning | Units |
|---|---|---|
| $K_d(T)$ | Dissociation rate (Arrhenius) | mol m$^{-2}$ s$^{-1}$ Pa$^{-1}$ |
| $K_r(T)$ | Recombination rate (Arrhenius) | m$^4$ mol$^{-1}$ s$^{-1}$ |
| $p_{\text{gas}}$ | Gas-phase partial pressure | Pa |

Both $K_d$ and $K_r$ follow the Arrhenius form (3). Setting $p_{\text{gas}} = 0$ models a vacuum or purge side (pure desorption).

*Equilibrium and Sieverts limit*

At equilibrium (zero net flux), $C = C_{\text{eq}}$:

$$K_d\,p_{\text{gas}} = K_r\,C_{\text{eq}}^2 \qquad \Rightarrow \qquad C_{\text{eq}} = \sqrt{\frac{K_d\,p_{\text{gas}}}{K_r}} \tag{16}$$

This is exactly Sieverts' law (4) with $K_s = \sqrt{K_d/K_r}$: *the surface recombination BC generalises the Sieverts equilibrium BC and recovers it in the fast-kinetics limit* [3].

*Damköhler number*

The Damköhler number compares the surface-kinetics rate to the bulk-diffusion rate:

**Damköhler number**

$$\mathrm{Da} = \frac{K_r\, C_{\mathrm{eq}}\, L}{D} \tag{17}$$

| Da | Limiting step | Surface concentration |
|---|---|---|
| $\mathrm{Da} \gg 1$ | Diffusion-limited | $C_s \approx C_{\mathrm{eq}}$ (Sieverts equilibrium) |
| $\mathrm{Da} \approx 1$ | Mixed control | $C_s = \frac{\sqrt{5}-1}{2}\, C_{\mathrm{eq}}$ (see below) |
| $\mathrm{Da} \ll 1$ | Kinetically limited | $C_s \ll C_{\mathrm{eq}}$ |

*Steady-state surface concentration*

At steady state, the Robin condition (15) must equal the permeation flux $D\, C_s/L$ (assuming zero concentration at the purge face):

$$K_r\, C_s^2 + \frac{D}{L}\, C_s - K_d\, p_{\mathrm{gas}} = 0 \tag{18}$$

The physical (positive) root is:

**Steady-state surface concentration (general Da)**

$$C_s = \frac{-D/L + \sqrt{(D/L)^2 + 4\, K_r\, K_d\, p_{\mathrm{gas}}}}{2\, K_r} \tag{19}$$

**Special case** $\mathrm{Da} = 1$ (with $K_d = K_r$, $p_{\mathrm{gas}} = 1$, $D/L = K_r\, C_{\mathrm{eq}} = K_r$):

$$C_s^2 + C_s - 1 = 0 \qquad \Rightarrow \qquad C_s = \tfrac{\sqrt{5}-1}{2} \approx 0.618 \text{ mol m}^{-3} \tag{20}$$

The golden-ratio conjugate appears again. The flux is reduced by 38.2 % relative to the Sieverts equilibrium limit:

$$\frac{J_{\mathrm{surf}}}{J_{\mathrm{ref}}} = \frac{C_s}{C_{\mathrm{eq}}} = \frac{\sqrt{5}-1}{2} \approx 0.618 \tag{21}$$

**Numerical treatment.** The $C^2$ term is Picard-frozen at the previous cell concentration, making the gradient a simple field evaluation each PIMPLE outer iteration. Convergence is guaranteed by the PIMPLE outer loop.

**Demonstrated by:** `case322`. Sub-case `sieverts_bc` provides the $\mathrm{Da} \to \infty$ reference ($J_{\mathrm{ref}}$); sub-case `surface_recombination` runs $\mathrm{Da} = 1$ and confirms $C_s = (\sqrt{5}-1)/2$ and the 38.2 % flux reduction.

## Performance Metrics

### Permeation flux and speciesFlux function object

The area-integrated molar permeation flux through a patch is:

**Surface molar flux**

$$J_{\text{patch}} = -\int_{\text{patch}} D\,\nabla C \cdot \hat{n}\,\mathrm{d}A \qquad [\,\text{mol s}^{-1}] \tag{22}$$

The `speciesFlux` function object evaluates this integral at every write time without modifying the solver. It outputs both the area-averaged flux density $[\,\text{mol m}^{-2}\ \text{s}^{-1}]$ and the total flux $[\,\text{mol s}^{-1}]$ to `postProcessing/<name>/<time>/speciesFlux.dat`.

The diffusivity field `D_<species>` is registered as `AUTO_WRITE` by the `speciesModel`, making it available to the function object at every write time.

**Dictionary syntax:**

```
functions
{
    wallPermeation
    {
        type        speciesFlux;
        libs        ("libspeciesPost.so");
        region      wall;
        species     C_H2;
        patches     (wall_to_hitec);
        writeControl writeTime;
    }
}
```

**Demonstrated by:** `case320` (permeation flux through SS316 wall into Hitec coolant), `case321`, `case322` (permeation flux at purge face).

### Permeation reduction factor

The permeation reduction factor (PRF) quantifies the effectiveness of a barrier coating:

**Permeation reduction factor**

$$\text{PRF} = \frac{J_{\text{without barrier}}}{J_{\text{with barrier}}} = \frac{C_0/R_{\text{total,bare}}}{C_0/R_{\text{total,coated}}} \tag{23}$$

For a two-layer composite (coating thickness $L_c$, substrate thickness $L_s$) with equal $K_s$ on both sides, the steady-state PRF can be approximated by the ratio of total diffusion resistances. High PRF arises when the coating has a much larger $K_s/D$ ratio (lower permeability) than the substrate.

The permeability of a material is $\Phi = K_s^2 D$ (in $[\,\text{mol m}^{-1}\ \text{s}^{-1}\ \text{Pa}^{-1}]$, the Richardson–Dushman formula for diatomic gases).

**Demonstrated by:** `case319`. A tungsten carbide (WC) coating on SS316 at $T = 1073$ K:

$$\text{PRF} \approx 200$$

confirmed by comparing the `with_coating` and `no_coating` sub-cases. The PRF is dominated by the large $K_s$ ratio ($K_{s,\text{WC}} \gg K_{s,\text{SS316}}$), which lowers the dissolved concentration in the coating and therefore reduces the flux into the substrate.

## Numerical Implementation

### Finite-volume discretisation

The species equation (1) is discretised using the standard OpenFOAM finite-volume method on an unstructured collocated mesh:

- **Time integration:** first-order implicit Euler (`ddtSchemes: Euler`).
- **Diffusion:** second-order Gauss linear with explicit non-orthogonal correction (`laplacianSchemes: Gauss linear corrected`).
- **Gradient:** Gauss linear (`gradSchemes: Gauss linear`).

These settings are appropriate for diffusion-dominated transport on orthogonal structured meshes (all tutorial cases use 1-D block meshes).

### PIMPLE outer loop and Picard convergence

The nonlinear boundary conditions (`quadratic`, `sqrt`, `surfaceRecombination`) are linearised by Picard (fixed-point) iteration within the PIMPLE outer corrector loop:

1. **Outer iteration $k$:** the nonlinear BC evaluates the interface condition using cell values from iteration $k-1$ (frozen coefficients).
2. **Species equation:** assembled and solved with the frozen coefficients; cell values are updated.
3. **Convergence check:** if the species residual $< \varepsilon$ (`residualControl`), the outer loop exits; otherwise another outer iteration is performed.

The mixed-BC weight (12) uses the *linearised derivative* of the partition function evaluated at the current iterate, ensuring that flux continuity is enforced at every Picard step and that the method converges to the exact partition condition as $k \to \infty$.

### Coupled-patch data exchange

At each outer PIMPLE iteration, the coupled patches exchange:

- **Neighbour cell concentration** $C_{n,\text{cell}}$ (used to evaluate the partition refValue),
- **Neighbour diffusivity** $D_n$ and **inverse distance** $\delta_n^{-1}$ (used to evaluate nbrKD for the weight).

Data exchange uses OpenFOAM's mapped-patch infrastructure with an incremented MPI message tag to avoid collisions when multiple coupled fields are exchanged in the same iteration.

### Semi-implicit trapping ODE

The McNabb–Foster ODE (6) is solved *cell by cell* with the per-cell semi-implicit Euler update (7). This is performed explicitly before the species equation assembly, so the sink term $\partial C_t/\partial t$ is available for the right-hand side. No global matrix inversion is required for the ODE update.

### Library architecture

| Library | Purpose | Key classes |
|---|---|---|
| `libspeciesTransport.so` | Core physics | `speciesModel`, `arrheniusProperty`, `McNabbFoster`, `noTrapping` |
| `libspeciesCoupling.so` | Boundary conditions | `sievertsCoupledMixed`, `surfaceRecombination` |
| `libspeciesSolid.so` | Solver module | `speciesSolid` |
| `libspeciesPost.so` | Post-processing | `speciesFlux` |

All libraries must be listed in `system/controlDict`:

```
libs ("libspeciesTransport.so" "libspeciesCoupling.so"
      "libspeciesSolid.so"    "libspeciesPost.so");
```

## Tutorial Guide

The twelve tutorial cases are ordered by increasing physical complexity. Each case contains an `Allrun` script, an `Allclean` script, a per-case `README.md`, and a Python validation script that compares the simulation output to an analytical or benchmark solution.

### Case summary table

### Physics-to-tutorial cross-reference

### Running the tutorials

**Prerequisites.** Source OpenFOAM-13 and build the libraries:

```
source /opt/openfoam13/etc/bashrc
cd $WM_PROJECT_USER_DIR/multiSpeciesRegionFoam
./Allwmake 2>&1 | tee build.log
```

**Individual case:**

```
cd tutorials/case314-composite-membrane
./Allrun
python3 validation/membrane_compare.py --plot
```

**Validation exit code:** each validation script exits 0 on PASS (flux within 5 % of analytical) and 1 on FAIL.

Table 1: Tutorial cases and the physics they demonstrate.

| Case | Solver | Physics / Key feature |
|---|---|---|
| `case311` | `solid` | Pure 1-D diffusion, step BC. Uses temperature field as species proxy. Validates erfc front propagation (exact analytical solution). |
| `case312` | `speciesSolid` | 1-D diffusion, non-uniform initial condition (`setFields`). Fourier cosine series analytical solution. First use of the species equation. |
| `case313` | `speciesSolid` | McNabb–Foster trapping (§3.3). Two sub-cases: weak trapping ($\alpha_t/D \ll 1$) and strong trapping ($\alpha_t/D \gg 1$). Validates semi-implicit ODE (7). |
| `case314` | `speciesSolid` | Two-region composite membrane; Sieverts–Sieverts linear partition (§4.2) with $K_{s,A}/K_{s,B} = 4$. Validates concentration jump at the interface. |
| `case315` | `speciesSolid` | Isothermal membrane distillation. Henry-law solubility (§3.2), constant $D$. Linear steady-state profile. |
| `case316` | `speciesSolid` | Membrane distillation with Arrhenius $D(T)$ (§3.1). Temperature gradient drives non-linear steady-state profile. Full energy–species coupling active. |
| `case317` | `speciesSolid` | Reverse osmosis: two-region feed/membrane. Henry partition at the feed face ($K_H = 0.5$). Solution-diffusion model. |
| `case318` | `speciesSolid` | Code-to-code benchmark against Pasler et al. and Fourier sine series. Constant BCs; transient diffusion with exact analytical solution. |
| `case319` | `speciesSolid` | WC coating + SS316 substrate; real Arrhenius $D(T)$ for both materials. Linear partition ((8)). PRF $\approx 200$ (§5.2). Two sub-cases: `with_coating` / `no_coating`. |
| `case320` | `speciesSolid` | Three-region shell-and-tube heat exchanger: FLiBe (shell) \| SS316 (wall) \| Hitec (tube). Full conjugate heat transfer + species permeation through the wall. Arrhenius $D(T)$ from $T$ gradient through wall. `speciesFlux` function object measures $J$ through the wall. |
| `case321` | `speciesSolid` | Henry–Sieverts partition (§4.3, §4.2). Two sub-cases: `sieverts_sieverts` (reference) and `henry_sieverts` (sqrt/quadratic BC). Validates 23.6 % flux increase and interface concentration jump. Golden-ratio interface concentration (13). |
| `case322` | `speciesSolid` | Surface recombination/dissociation (§4.4). Two sub-cases: `sieverts_bc` (Da $\to \infty$ reference) and `surface_recombination` (Da = 1). Validates golden-ratio surface concentration (20) and 38.2 % flux reduction. |

Table 2: Which tutorials exercise which physical models.

| Physical model | *311* | *312* | *313* | *314* | *315* | *316* | *317* | *318* | *319* | *320* | *321* | *322* |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Species diffusion eq. (1) | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Energy eq. (2) | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Arrhenius (3) | | | | | | ✓ | | | ✓ | ✓ | | |
| Sieverts law (4) | | | ✓ | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ |
| Henry law (5) | | | | | ✓ | ✓ | ✓ | | | | ✓ | |
| McNabb–Foster (6) | | | ✓ | | | | | | | | | |
| Linear partition (8) | | | | ✓ | | | | ✓ | ✓ | ✓ | ✓ | |
| Sqrt partition (10) | | | | | | | | | | | ✓ | |
| Quadratic partition (11) | | | | | | | ✓ | | | | ✓ | |
| Surface recombination (15) | | | | | | | | | | | | ✓ |
| PRF metric (23) | | | | | | | | | ✓ | | | |
| speciesFlux FO (22) | | | | | | | | | | ✓ | ✓ | ✓ |
| Multi-region ($\geq 3$ regions) | | | | | | | | | | ✓ | | |

## Notation Summary

| Symbol | Meaning | SI units |
|---|---|---|
| $C$ | Mobile species concentration | mol m$^{-3}$ |
| $C_t$ | Trapped concentration | mol m$^{-3}$ |
| $D(T)$ | Diffusivity (Arrhenius) | m$^2$ s$^{-1}$ |
| $E_a$ | Activation energy | J mol$^{-1}$ |
| $f_t$ | Trap fraction | — |
| $J$ | Molar flux (area-averaged) | mol m$^{-2}$ s$^{-1}$ |
| $K_d(T)$ | Dissociation rate | mol m$^{-2}$ s$^{-1}$ Pa$^{-1}$ |
| $K_H(T)$ | Henry solubility constant | mol m$^{-3}$ Pa$^{-1}$ |
| $K_r(T)$ | Recombination rate | m$^4$ mol$^{-1}$ s$^{-1}$ |
| $K_s(T)$ | Sieverts solubility constant | mol m$^{-3}$ Pa$^{-1/2}$ |
| $L$ | Layer thickness | m |
| $N$ | Molar trap-site density | mol m$^{-3}$ |
| $n_t = f_t N$ | Available trap density | mol m$^{-3}$ |
| $p$ | Gas-phase partial pressure | Pa |
| $R$ | Gas constant (8.314) | J mol$^{-1}$ K$^{-1}$ |
| $S_{\mathrm{vol}}$ | Volumetric species source | mol m$^{-3}$ s$^{-1}$ |
| $T$ | Temperature | K |
| $w$ | Mixed-BC value fraction | — |
| $X_0$ | Arrhenius pre-exponential factor | (same as $X$) |
| $\alpha_d(T)$ | Detrapping rate | s$^{-1}$ |
| $\alpha_t$ | Trapping rate | s$^{-1}$ |
| $\delta^{-1}$ | Cell-centre to face distance (inv.) | m$^{-1}$ |
| $\varepsilon$ | Trap binding energy | eV |
| $\kappa(T)$ | Thermal conductivity | W m$^{-1}$ K$^{-1}$ |
| $\Phi = K_s^2 D$ | Permeability (diatomic gas) | mol m$^{-1}$ s$^{-1}$ Pa$^{-1}$ |
| Da | Damköhler number (surface kinetics) | — |
| PRF | Permeation reduction factor | — |

## References


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